Supplementary Material

Towards Efficient LLM-Based Code Generation: A Systematic Review

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Overview

This supplement carries the per-cell experimental detail that the main text summarizes. Section 1 documents the generation, judging, and energy protocol; Sections 2–5 reproduce the composition, Pareto, cross-platform, acceptance, cross-benchmark, and second-family tables. All values come from the same runs and the same replication package as the main text, and the numbering of main-text tables is unchanged. Table numbers here are cited as Table S1–S8 from the main text.

1 Experimental protocol

All main experiments use the Instruct checkpoints of the Qwen2.5-Coder family at six scales (0.5B, 1.5B, 3B, 7B, 14B, 32B). Concretely, these are the Qwen2.5-Coder Instruct checkpoints at 0.5B, 1.5B, 3B, 7B, 14B, and 32B; the draft is the 0.5B-Instruct checkpoint, and the second family is DeepSeek-Coder-6.7B-Instruct. The prompt is the raw benchmark prompt without a chat template or system message, except for the LiveCodeBench slice and the BigCodeBench instruct cells, which use the official chat protocol of their benchmark. Greedy decoding is deterministic up to floating-point kernel order. Adaptive sampling draws up to ten completions at temperature 0.8 and stops once the observed pass rate reaches 0.95 after at least three samples; its any-pass rate and cumulative latency are compared only within the adaptive group. Generation is capped at 512 new tokens. The parser truncates a completion at the first non-indented, non-comment, non-decorator top-level line after the first. The judge runs the prompt, completion, and reference tests in a subprocess with a five-second timeout on HumanEval and MBPP, twenty seconds on BigCodeBench, and six seconds per test inside the official LiveCodeBench judge. The BigCodeBench judge additionally enforces a 16 GB address-space limit and a CPU-time limit of eight times the wall-clock timeout. Two warm-up generations precede each configuration, so absolute pass@1 values are harness-relative.

Energy is measured with `nvmlDeviceGetTotalEnergyConsumption` before and after each request, with a preceding `torch.cuda.synchronize`. The boundary is the GPU board; CPU and DRAM energy are excluded. Energy values come from one dedicated full-benchmark round per configuration, except the three V100 speculative-adaptive cells, whose energy is the mean over the composition runs that captured NVML readings. Peak memory is the minimum across the runs of a cell, the reading taken from a fresh allocation state. Pass@1, latency, and throughput are means over five runs on A800 and three on V100, with standard deviations.

The five hypotheses and their analysis plan were registered on 2026-08-31, before any A800 result was inspected. The plan fixes median wall-clock time per request as the primary latency metric, a significance level of 0.05 with Holm-Bonferroni correction, paired tests with the run as the pairing unit, and the reporting of rejected hypotheses as negative results. Three deviations apply: the V100 server left the group in July 2026, so all five composition and Pareto runs executed on A800 and only run-level V100 aggregates survive; the registered 32B full-precision cell does not fit on one 40 GB GPU; and round-level pairing with $n = 5$ replaced the registered task-level pairing because the aggregated files

Table 1: A800 replication of the composition matrix on HumanEval, Qwen2.5-Coder-7B-Instruct, five runs per cell. Latency is the median wall-clock time per request, memory is the minimum across runs, and energy is the single-pass NVML measurement. Adaptive rows report the any-pass rate and cumulative latency over up to ten samples and are comparable only with each other.

Precision	Decoding	Sampling	pass@1 (% ± SD)	p50 Latency (ms ± SD)	tok/s (± SD)	Memory (MB)	Energy (J/req)
FP16	Standard	Greedy	81.71 ± 0.00	3139 ± 62	44.3 ± 0.8	14609	698
INT8	Standard	Greedy	80.49 ± 0.00	17501 ± 268	8.2 ± 0.1	8680	1382
INT4	Standard	Greedy	80.49 ± 0.00	4996 ± 82	27.2 ± 0.5	5787	583
FP16	Speculative	Greedy	81.10 ± 0.00	5873 ± 118	25.1 ± 0.5	15572	556
INT8	Speculative	Greedy	80.49 ± 0.00	9411 ± 151	16.7 ± 0.3	9632	757
INT4	Speculative	Greedy	80.49 ± 0.00	6334 ± 115	22.4 ± 0.4	6755	634
FP16	Standard	Adaptive	93.90 ± 1.14	16607 ± 907	43.4 ± 0.8	14609	4328
INT8	Standard	Adaptive	93.90 ± 0.75	89811 ± 4953	8.2 ± 0.1	8680	8957
INT4	Standard	Adaptive	94.02 ± 0.80	28434 ± 2239	26.7 ± 0.5	5787	3847
FP16	Speculative	Adaptive	93.90 ± 1.22	27516 ± 1680	26.9 ± 0.7	15612	3073
INT8	Speculative	Adaptive	93.17 ± 1.00	39585 ± 2732	19.0 ± 0.2	9639	3619
INT4	Speculative	Adaptive	93.41 ± 0.80	30818 ± 2449	24.7 ± 0.6	6772	3543

Table 2: Latency speedup over the FP16 standard-decoding greedy-sampling baseline, V100 runs. A800 counterparts appear in Table S5.

Configuration	Observed	Predicted
INT8	0.23×	—
INT4	0.63×	—
FP16, speculative	0.50×	—
INT8, speculative	0.35×	0.12×
INT4, speculative	0.43×	0.32×

of runs 1–4 lack per-problem records. The A800 suite comprises 183 multi-run benchmark runs; the second-family, acceptance, instruct-protocol, and HumanEval+ cells are single runs per configuration. The original V100 campaign contributes 81 runs.

Hardware: V100-SXM2-32GB (4 GPUs, 226 GB RAM, PyTorch 2.5.1/CUDA 12.4); A800-SXM4-40GB (two servers, 2 GPUs each, Xeon 8358P, 196 GB RAM, PyTorch 2.6.0/CUDA 12.4, transformers 5.16.1, bitsandbytes 0.50.2); RTX 3090 Ti (2 GPUs, 24 GB each, Ryzen 9 5950X, 125 GB RAM, PyTorch 2.5.1, transformers 5.9.0, bitsandbytes 0.49.2, no board-energy counter); and a laboratory V100-PCIE-32GB server (2 GPUs, 32 cores, driver 550.54.15, PyTorch 2.6.0, transformers 5.5.4, bitsandbytes 0.49.2) for LiveCodeBench, the acceptance loop, the two 32B cells, and the BigCodeBench instruct cells. Quantization uses bitsandbytes: INT8 via `load_in_8bit` and INT4 via NF4 with FP16 compute. Speculative decoding uses the HuggingFace `assistant_model` API; the Qwen draft is Qwen2.5-Coder-0.5B-Instruct in FP16, and the second-family draft is DeepSeek-Coder-1.3B-Instruct.

2 Composition results

Table S1 reports the five-run A800 composition replication. Table S2 reports the latency speedups over the FP16 standard-decoding greedy baseline and the multiplicative prediction.

Table 3: V100 results. Pareto frontier results: 5 model scales × 3 precisions on HumanEval under greedy decoding, averaged over 3 runs on V100, with standard deviations for pass@1, latency, and throughput. Memory is the minimum across runs, as the main text describes. Energy is the single-pass measurement, and per-run values ship with the replication package.

Model	Precision	pass@1 (%)	Latency (ms)	tok/s	Memory (MB)	Energy (J/req)
0.5B	FP16	45.1 ± 0.0	9207 ± 937	29.8 ± 2.8	992	563
0.5B	INT8	43.9 ± 0.0	54568 ± 12195	5.4 ± 1.1	651	2383
0.5B	INT4	40.9 ± 0.0	17974 ± 102	18.7 ± 0.2	501	1104
1.5B	FP16	46.3 ± 0.0	10124 ± 601	26.3 ± 1.3	3003	751
1.5B	INT8	48.2 ± 0.0	53669 ± 9443	4.8 ± 0.9	1803	2613
1.5B	INT4	51.8 ± 0.0	16190 ± 265	15.8 ± 0.3	1250	1057
3B	FP16	74.4 ± 0.0	11340 ± 223	21.0 ± 0.4	5959	1027
3B	INT8	72.0 ± 0.0	64314 ± 16537	4.0 ± 0.9	3339	3400
3B	INT4	72.6 ± 0.0	17077 ± 417	12.6 ± 0.3	2228	1332
7B	FP16	81.7 ± 0.0	6583 ± 236	25.1 ± 0.8	14644	986
7B	INT8	79.3 ± 0.0	37453 ± 15074	5.1 ± 1.5	8685	2058
7B	INT4	80.5 ± 0.0	11103 ± 1284	14.7 ± 1.6	5914	1089
14B	FP16	86.6 ± 0.0	14981 ± 2197	13.9 ± 1.9	28412	2369
14B	INT8	84.1 ± 0.0	59055 ± 5985	3.4 ± 0.3	16066	3889
14B	INT4	88.4 ± 0.0	22273 ± 428	9.2 ± 0.2	10472	2361

Table 4: The two large-scale speculative cells on the A800, two runs each, against their standard-decoding counterparts of the same scale and precision.

Scale	Precision	pass@1 (%)	p50 (ms)	E (J)	vs standard
14B	FP16	87.20	11863	1110	1.47× slower
32B	INT4	83.54	15210	2239	1.28× faster, −27.3% E

3 Pareto frontier detail

Table S3 reports the full V100 Pareto frontier. Table S4 reports the two large-scale speculative cells on the A800 against their standard-decoding counterparts.

4 Cross-platform and acceptance detail

Table S5 reports the cross-platform replication, including the RTX 3090 Ti and HumanEval+ columns. Table S6 reports the instrumented draft-verify loop.

5 Generalization detail

Table S7 reports the LiveCodeBench layer with per-difficulty breakdowns. Table S8 reports the second-family check at 7B.

Table 5: Cross-architecture comparison on HumanEval, Qwen2.5-Coder-7B-Instruct. Latency ratios are relative to the FP16 standard greedy baseline of the same platform. The V100 column uses run-level means without per-task records. The A800 column is raw data over five runs with per-cell median latency. The RTX 3090 Ti column is a single exploratory run under a third software stack described in the main text. The HumanEval+ rows report the EvalPlus judge over a regeneration pass on each platform. On the HumanEval+ rows, V100 and A800 agree exactly in three of the four cells and within one task in the fourth; the base pass@1 rows agree within two tasks in every cell, the INT8 cell being the widest, and the 3090 Ti agrees with the A800 within one task in all four cells of both layers. Board energy is not measurable on the consumer card. The V100 latency column is the mean of per-request means of the original campaign, whereas the A800 and 3090 Ti columns report medians, so the latency rows mix two estimators.

Measure	**V100, 3 runs**	**A800, 5 runs**	**3090 Ti, 1 run**
FP16 standard, pass@1	81.7%	81.71%	81.10%
FP16 standard, latency	6,562 ms	3,139 ms	2,923 ms
INT8 standard, pass@1	79.3%	80.49%	80.49%
INT8 standard, latency ratio	4.38×	5.58×	4.50×
INT4 standard, latency ratio	1.58×	1.59×	1.45×
FP16 speculative, latency ratio	1.98×	1.87×	1.51×
INT8 speculative, observed vs predicted	0.35× vs 0.12×	0.33× vs 0.10×	0.46× vs 0.15×
INT4 speculative, observed vs predicted	0.43× vs 0.32×	0.50× vs 0.34×	0.51× vs 0.46×
FP16 standard, HumanEval+ pass@1	76.22%	76.22%	76.22%
INT8 standard, HumanEval+ pass@1	73.78%	74.39%	75.00%
INT4 standard, HumanEval+ pass@1	75.61%	75.61%	75.00%
FP16 speculative, HumanEval+ pass@1	76.22%	76.22%	76.22%
FP16 standard, energy per request	1,174 J	698 J	—

Table 6: Draft acceptance rates measured with an instrumented draft-verify loop on a fixed 60-problem HumanEval slice, draft model Qwen2.5-Coder-0.5B-Instruct in FP16, five proposed tokens per round. τ is the mean number of effective tokens per verification round including the bonus token. The A800 columns are the primary measurement. The RTX 3090 Ti and V100 acceptance columns repeat the same loop once, the latter on the laboratory V100 server, and neither covers 32B. V100 net latency comes from the three-run campaign at 7B and from the single-run replication cells at 32B. Net latency at 7B is relative to the FP16 standard-decoding cell of the same platform; at 32B it is relative to the same-precision standard cell.

	A800			**RTX 3090 Ti**		**V100**	
Target	**Acc.**	τ	**Net latency**	**Acc.**	**Net**	**Acc.**	**Net**
7B FP16	0.810	4.78 ± 0.66	1.87× slower	0.809	1.51× slower	0.808	1.98× slower
7B INT8	0.798	4.72 ± 0.69	3.00× slower	0.809	2.17× slower	0.803	2.85× slower
7B INT4	0.781	4.80 ± 0.53	2.02× slower	0.779	1.94× slower	0.782	2.31× slower
32B INT4	0.707	4.50 ± 0.55	1.28× faster	—	—	—	1.21× slower

Table 7: Contamination-free layer: LiveCodeBench codegeneration, 288 problems released from 2024-11-02 to 2025-04-06, after the release of every model evaluated on this layer. Qwen2.5-Coder-7B-Instruct under the official chat-template prompt and judge over public and private tests, one run per cell, 512-token budget, on the laboratory V100 and the RTX 3090 Ti. Ratios are p50 latency relative to the FP16 standard cell of the same machine. J/req is NVML board energy per request, not measurable on the consumer card. The V100 cells exhaust the budget on 7, 9, 5, and 7 problems and the 3090 Ti cells on 8, 10, 7, and 6 problems respectively.

Config	pass@1 (%)	Easy	Med	Hard	p50 (ms)	Ratio	J/req
Tesla V100-PCIE-32GB (laboratory server), 1 run							
FP16 std	17.36	51.35	10.23	2.38	8,197	1.00	1,838
INT8 std	15.97	50.00	6.82	2.38	34,618	4.22	2,112
INT4 std	18.75	56.76	10.23	2.38	13,888	1.69	2,144
FP16 spec	17.71	52.70	10.23	2.38	21,520	2.63	3,175
GeForce RTX 3090 Ti, 1 run							
FP16 std	17.71	52.70	10.23	2.38	4,593	1.00	—
INT8 std	17.71	52.70	9.09	3.17	16,526	3.60	—
INT4 std	18.75	59.46	7.95	2.38	6,510	1.42	—
FP16 spec	17.36	51.35	10.23	2.38	9,589	2.09	—

Table 8: Second-family check at 7B: DeepSeek-Coder-6.7B-Instruct on the A800 under greedy decoding, one run per cell, with the same-family 1.3B-Instruct draft in the speculative cell. Ratios are p50 latency relative to the FP16 standard cell of the same family, with the Qwen ratios of the main-text cross-benchmark table for comparison. Every DeepSeek completion exhausts the 512-token budget, so only within-family ratios are comparable.

Config	pass@1 (%)	HE+ (%)	p50 (ms)	Ratio	Qwen ratio
FP16 std	69.51	64.63	12,146	1.00	1.00
INT8 std	70.73	65.24	71,793	5.91	5.58
INT4 std	67.68	61.59	22,336	1.84	1.59
FP16 spec	69.51	64.63	12,399	1.02	1.87